



Ext 2

Number of Pages Used _____

NORTH SYDNEY BOYS HIGH SCHOOL

2009
ASSESSMENT TASK 3

Mathematics

Extension 2

Examiner: GB

General Instructions

- Working time – 60 minutes
- Write in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Barrett
- ☐ Mr Fletcher
- ☐ Mr Weiss

Student Number:

Question No	1	2	3	4	5	6	Total	Total
Mark	$\frac{17}{17}$	$\frac{3}{3}$	$\frac{10}{10}$	$\frac{5}{5}$	$\frac{8}{8}$	$\frac{5}{5}$	$\frac{48}{48}$	$\frac{100}{100}$

(To be used by the exam markers only.)

Question 1 (17 marks)**Marks**

Integrate the following

a) $\int \frac{dx}{x(\log_e x)^2}$ 2

b) $\int \frac{x+2}{x^2+6} dx$ 2

c) $\int \frac{e^x}{\sqrt{e^{2x}+1}} dx$ 3

d) $\int_0^{\frac{\pi}{4}} x \sin 2x \, dx$ 3

e) $\int \frac{3x^2+10x-5}{(x+1)^2(x-2)} dx$ Use partial fractions 4

f) $\int \frac{dx}{\sin x + \tan x}$ 3

Question 2 (3 Marks)

Solve: $\sin x - \cos x = \frac{\sqrt{6}}{2}$ $0 \leq x \leq 2\pi$

Leave your answer in exact form.

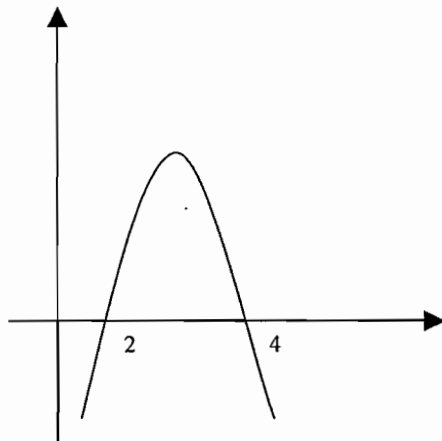
Continued on page 2

Question 3 (10 Marks)

The diagram below shows the graph of the function $y = -x^2 + 6x - 8$.

On separate diagrams and without using calculus, sketch the following graphs.

Indicate clearly any asymptotes and intercepts with the axes.



- | | | |
|----|----------------------|---|
| a) | $y = f(x) $ | 2 |
| b) | $y^2 = f(x)$ | 3 |
| c) | $y = \frac{1}{f(x)}$ | 3 |
| d) | $y = e^{f(x)}$ | 2 |

Continued on page 3

Question 4 (5 Marks)

If $I_n = \int x(\ln x)^n dx$

a) Prove that $I_n = \frac{x^2(\ln x)^n}{2} - \frac{n}{2}I_{n-1}$ 2

b) Hence find $\int_1^3 x(\ln x)^2 dx$ 3

Question 5 (8 Marks)

Sketch the following curves on separate number planes, showing all their important features.

a) $y^2 = (x - 1)(x - 3)^2$ 3

b) $|y| = 16 - x^2$ 2

c) $y = \sin(\cos^{-1} x)$ 3

Question 6 (5 Marks)

The area bounded by the curve $y = 4 - x^2$ and the x axis is rotated about the line $x = 2$. Use the method of slices to find the volume of the solid formed. Show all your working.

END OF EXAMINATION

Q1 a $\int \frac{1}{x} \times (\log_e x)^2 dx$

$$= -1 \times (\log_e x)^{-1} + C$$

$$= -\frac{1}{\log_e x} + C$$

1/2

b) $\int \frac{x+2}{x^2+6} dx = \int \frac{x}{x^2+6} dx + \int \frac{2}{x^2+6} dx$

$$= \frac{1}{2} \log_e(x^2+6) + \frac{2}{\sqrt{6}} \tan^{-1}\left(\frac{x}{\sqrt{6}}\right) + C$$

1

1/2

c) $\int \frac{e^x}{\sqrt{e^{2x}+1}} dx$ let $u = e^x$
 $du = e^x dx$

1

$$\Rightarrow \int \frac{du}{\sqrt{u^2+1}} = \ln(u + \sqrt{u^2+1}) + C$$

1

$$= \ln(e^x + \sqrt{e^{2x}+1}) + C$$

1/3

d) $\int_0^{\pi/4} x \sin 2x dx$

$$= \left[x \times -\frac{1}{2} \cos 2x \right]_0^{\pi/4} - \int_0^{\pi/4} \left[-\frac{1}{2} \cos 2x \right] dx$$

1

$$= 0 + \left[\frac{1}{4} \sin 2x \right]_0^{\pi/4}$$

1

$$= \frac{1}{4}$$

1/3

$$e) \int \frac{3x^2 + 10x - 5}{(x+1)^2(x-2)} dx.$$

$$\int \frac{ax+b}{(x+1)^2} + \frac{c}{x-2} dx$$

$$\Rightarrow (ax+b)(x-2) + c(x+1)^2 = 3x^2 + 10x - 5$$

$$(a+c) = 3 \quad (1)$$

$$\text{let } x = 2 \Rightarrow \begin{aligned} 9c &= 27 \\ c &= 3 \quad (2) \end{aligned}$$

$$(2) \text{ into } (1) \Rightarrow \underline{a = 0}$$

$$\begin{aligned} x = 0 &\Rightarrow -2b + c = -5 \\ -2b &= -8 \\ \underline{b} &= \underline{4} \end{aligned}$$

$$\int \left(\frac{4}{(x+1)^2} + \frac{3}{x-2} \right) dx$$

$$= \frac{-4}{x+1} + 3 \ln(x-2) + C$$

$$b1/4$$

$$f) \int \frac{dx}{\sin x + \tan x}$$

$$\text{let } t = \tan \frac{x}{2} \quad \therefore dx = \frac{2dt}{1+t^2}$$

$$\Rightarrow \int \frac{\frac{2dt}{1+t^2}}{\frac{2t}{1+t^2} + \frac{2t}{1-t^2}}$$

$$= \int \frac{\frac{dt}{1+t^2}}{\frac{t(1-t^2) + t(1+t^2)}{(1+t^2)(1-t^2)}}$$

$$= \int \frac{1-t^2 dt}{2t}$$

$$= \int \left(\frac{1}{2t} - \frac{t}{2} \right) dt$$

$$= \frac{1}{2} \ln(\tan \frac{x}{2}) - \frac{1}{4} \left(\tan \frac{x}{2} \right)^2 + C$$

$$1/3$$

$$= \boxed{17}$$

Q2

$$\sin x - \cos x = \frac{\sqrt{6}}{2}$$

$$\text{Let } A \sin(x - \alpha) = \sin x - \cos x = \frac{\sqrt{6}}{2}$$

$$A \cos \alpha = 1$$

$$A \sin \alpha = 1$$

$$\therefore \tan \alpha = 1 \Rightarrow \alpha = \frac{\pi}{4}$$

$$A^2 \cos^2 \alpha + A^2 \sin^2 \alpha = A^2$$

$$A^2 = 1 + 1$$

$$\therefore A = \sqrt{2}$$

$$\sqrt{2} \sin\left(x - \frac{\pi}{4}\right) = \frac{\sqrt{6}}{2}$$

$$\sin\left(x - \frac{\pi}{4}\right) = \frac{\sqrt{3}}{2}$$

$$x - \frac{\pi}{4} = \frac{\pi}{3}$$

or

$$= \frac{2\pi}{3}$$

$$\therefore x = \frac{7\pi}{12} \text{ or } \frac{11\pi}{12}$$

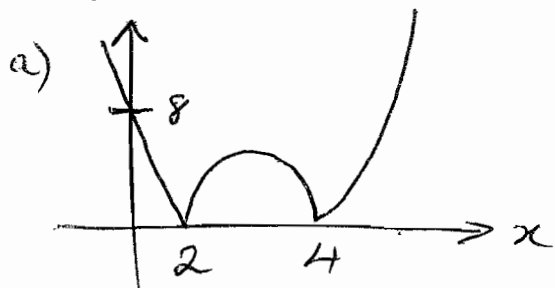
1 method.

1

1/3

3

Q3

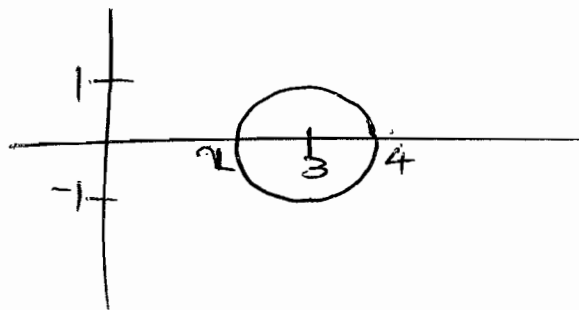


b)

$$y^2 = -x^2 + 6x - 8$$

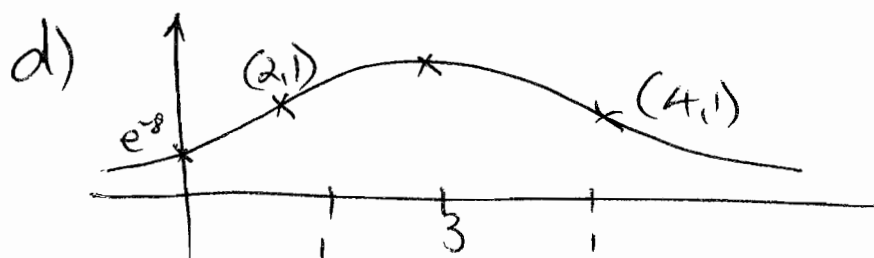
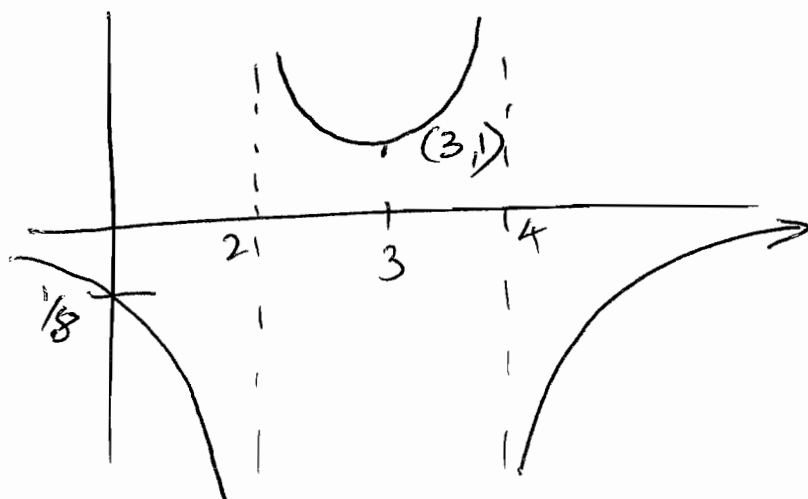
$$x^2 + y^2 - 6x = -8$$

$$(x-3)^2 + y^2 = 1$$



c)

$$y = \frac{-1}{x^2 - 6x + 8} = \frac{-1}{(x-2)(x-4)}$$



1 for y intercept = 8

1 for correct shape

/2

1 for circle
2 for intercepts

/3

1 Shape
1 Asymptotes
1 intercepts etc

/3

1 - Shape

1 - Nos

/2

10

Q4

$$a) I_n = \int x (\ln x)^n dx$$

$$= \left[\frac{x^2}{2} (\ln x)^n \right] - \frac{n}{2} \int x^2 \times \frac{1}{x} (\ln x)^{n-1} dx$$

$$= x^2 \frac{(\ln x)^n}{2} - \frac{n}{2} \int x (\ln x)^{n-1} dx$$

$$I_n = x^2 \frac{(\ln x)^n}{2} - \frac{n}{2} I_{n-1}$$

$$b) \int_1^3 x (\ln x)^2 dx = \left[\frac{x^2 \ln x^2}{2} \right]_1^3 - 1 \times \int_1^3 x \ln x dx$$

$$= \frac{9(\ln 3)^2}{2} - 0 - \left\{ \left[\frac{x^2 \ln x}{2} \right]_1^3 - \frac{1}{2} \int_1^3 x (\ln x)^0 dx \right\}$$

$$= \frac{9(\ln 3)^2}{2} - \frac{9 \ln 3}{2} + \frac{1}{2} \left[\frac{x^2}{2} \right]_1^3$$

$$= \frac{9(\ln 3)^2}{2} - \frac{9 \ln 3}{2} + 2$$

$$= \frac{9(\ln 3)}{2} [\ln 3 - 1] + 2$$

1 method shown.

must have
the expanded form
showing understanding
of the use of 'parts'

1

~~2~~

1

1

1/3

5

Q5

a) $y^2 = (x-1)(x-3)^2$

$$2y \frac{dy}{dx} = (x-1) \times 2(x-3) + (x-3)^2 \times 1$$

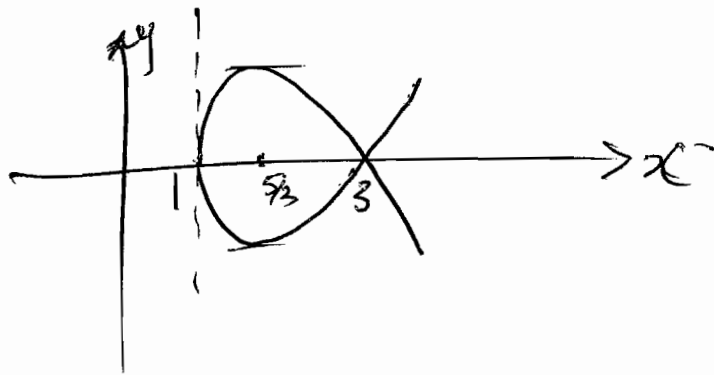
$$= (x-3)(2(x-1) + (x-3))$$

$$= (x-3)(3x-5)$$

$$\therefore \frac{dy}{dx} = \frac{(x-3)(3x-5)}{(x-3)^2(x-1)}$$

$\Rightarrow$ Vertical tangent at $x=-1$

$\Rightarrow$ critical point at $x=3$



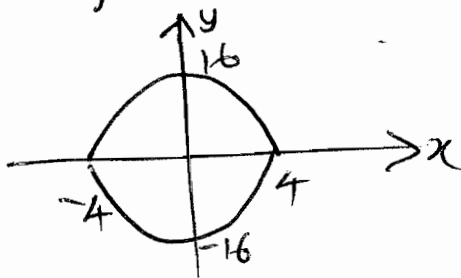
or

$D: x \geq 1$

$y = \pm (x-1)\sqrt{x-1}$ etc

b) $|y| = |6-x^2|$

$D: -4 \leq x \leq 4$



(Note! 2 parabolas a
Not rounded at ± 4

1 shape

1 intercepts

2.

c) $y = \sin(\cos^{-1}x)$

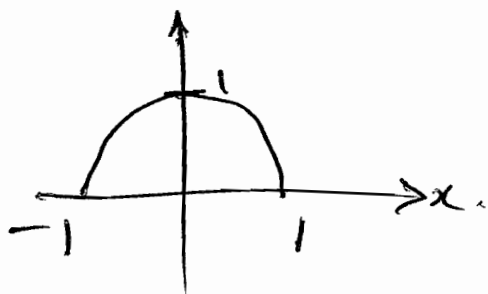
$\cos^{-1}x = \alpha \Rightarrow y = \sin \alpha.$

$\Rightarrow x = \cos \alpha$

$x^2 + y^2 = \sin^2 \alpha + \cos^2 \alpha = 1$

D: $-1 \leq x \leq 1$

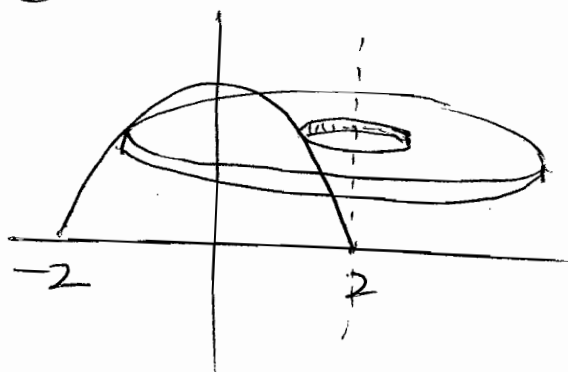
R: $0 \leq y \leq 1$



$1/3$

$\boxed{8}$

Q6



$y = 4 - x^2$

$x^2 = 4 - y$

$x = \sqrt{4 - y}$

$\delta V = [\pi(2+x)^2 - \pi(2-x)^2] \delta y$

$\delta V = \pi [2+x - (2-x)] [2+x + 2-x] \delta y$

$= \pi \times 2x \times 4 \delta y$

$= 8\pi x \delta y = 8\pi \sqrt{4-x} \delta y$

$V = \lim_{\delta y \rightarrow 0} \sum_{y=0}^4 8\pi \sqrt{4-x} \delta y$

$= 8\pi \int_0^4 \sqrt{4-x} dx$

$= 8\pi \left[-\frac{2}{3} (4-y)^{3/4} \right]_0^4$

$= 8\pi \left[0 + \frac{2}{3} 4^{3/2} \right]$

$= \frac{128\pi}{3} u^3$

$\frac{1}{5}$

$\boxed{5}$